

Part I

SUMMARY of Sigma Notation and Riemann Sums

Summary

February 21, 2026

Useful formulas: $(1) \sum_{k=1}^n C = n \cdot C; \quad (2) \sum_{k=1}^n k = \frac{n(n+1)}{2}; \quad (3) \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \quad (3) \sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}$

Useful example of application of rules for sums:

$$\sum_{k=1}^n (ak + b) = \sum_{k=1}^n ak + \sum_{k=1}^n b = a \sum_{k=1}^n k + \sum_{k=1}^n b = a \frac{n(n+1)}{2} + nb, \text{ where } a \text{ and } b \text{ constants.}$$

SUMMARY of Riemann Sums:

Riemann sum: $\sum_{k=1}^n f(x_k^*) \Delta x,$

Right Riemann sum: $x_k^* = x_k;$ Left Riemann sum: $x_k^* = x_{k-1};$ Midpoint Riemann sum: $x_k^* = \frac{x_{k-1} + x_k}{2}.$

Width of each of n rectangles on the interval $[a, b]: \quad \Delta x = \frac{b-a}{n}$

Grid points for interval $[a, b]:$

$$x_0 = a,$$

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$$x_{k-1} = a + (k-1)\Delta x = a + (k-1)\frac{b-a}{n},$$

$$x_k = a + k\Delta x = a + k\frac{b-a}{n},$$

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$$x_n = a + n\Delta x = a + n\frac{b-a}{n} = b$$

Part II

Recitation Questions

Problem 1

February 21, 2026

Problem 1 Evaluate the sum.

$$(a) \sum_{k=1}^n 5$$

Explanation. $\sum_{k=1}^n 5 = n \cdot 5$

$$(b) \sum_{k=1}^{10} 5$$

Explanation. Using the previous result for $n = 10$, we get $\sum_{k=1}^{10} 5 = 10 \cdot 5 =$

$$50$$

or

$$\sum_{k=1}^{10} 5 = 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 = 50$$

$$(c) \sum_{k=1}^n 8k$$

Explanation. $\sum_{k=1}^n 8k = 8 \sum_{k=1}^n k = 8 \frac{n(n+1)}{2} = 4n(n+1)$

$$(d) \sum_{k=1}^4 8k$$

Explanation. Using the previous result for $n = 4$, we get $\sum_{k=1}^4 8k =$

$$4(4)(4+1) = 80$$

or

$$\sum_{k=1}^4 8k = 8(1) + 8(2) + 8(3) + 8(4) = 8 + 16 + 24 + 32 = 80$$

$$(e) \sum_{k=1}^n (6k^2 - 8k - 5)$$

Explanation.
$$\sum_{k=1}^n (6k^2 - 8k - 5) = \sum_{k=1}^n 6k^2 + \sum_{k=1}^n (-8k) + \sum_{k=1}^n (-5) =$$

$$= 6 \sum_{k=1}^n k^2 - 8 \sum_{k=1}^n k - \sum_{k=1}^n 5 =$$

Now we use the formulas.

$$= 6 \frac{n(n+1)(2n+1)}{6} - 8 \frac{n(n+1)}{2} - \sum_{k=1}^n 5 = n(n+1)(2n+1) - 4n(n+1) -$$

$$5n =$$

$$= 2n^3 - n^2 - 8n$$

(f)
$$\sum_{k=0}^2 \cos\left(\frac{\pi}{2}k\right)$$

Explanation.

$$\begin{aligned} \sum_{k=0}^2 \cos\left(\frac{\pi}{2}k\right) &= \cos\left(\frac{\pi}{2}(0)\right) + \cos\left(\frac{\pi}{2}(1)\right) + \cos\left(\frac{\pi}{2}(2)\right) \\ &= \cos(0) + \cos\left(\frac{\pi}{2}\right) + \cos(\pi) \\ &= 1 + 0 + -1 = 0. \end{aligned}$$

Problem 2

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Problem 2 (a) *If a function is positive and decreasing on an interval $[a, b]$, will a right Riemann sum underestimate or overestimate the area of the region under the graph of the function? Justify your answer.*

Explanation. Because the function is decreasing, a right Riemann sum will be an underestimate. By definition, $f(x_k) < f(x)$ for all x in the interval $[x_{k-1}, x_k]$.

(b) *If a function is positive and decreasing on an interval $[a, b]$, will a left Riemann sum underestimate or overestimate the area of the region under the graph of the function? Justify your answer.*

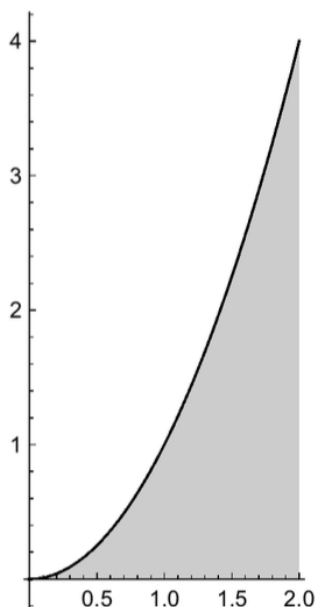
Explanation. Because the function is decreasing, a left Riemann sum will be an overestimate. By definition, $f(x_{k-1}) > f(x)$ for all x in the interval $[x_{k-1}, x_k]$.

Problem 3

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Problem 3

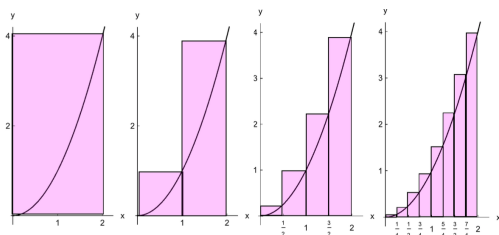
Problem 3 The graph of the function $f(x) = x^2$ is given in the figure.



Will a right Riemann sum approximation, for any value of n , be an underestimate or overestimate?

Explanation. Since the function f is positive and increasing on this interval, the right Riemann sum will always give an overestimate of the actual area.

Approximate the shaded area using a right Riemann sum with $n = 1, 2, 4$, and 8 rectangles, as illustrated in the figure below.



- (a) Approximate the shaded area using a right Riemann sum with $n = 1$ rectangles.

Explanation. $\Delta x = \frac{2}{1} = 2$
 $x_1^* = x_1 = 2$

$$A \approx f(2)\Delta x = 4(2) = 8$$

- (b) Approximate the shaded area using a right Riemann sum with $n = 2$ rectangles.

Explanation. $\Delta x = \frac{2}{2} = 1$

$$x_k^* = x_k = k\Delta x = k(1), \text{ for } k = 1, 2$$

$$A \approx \sum_{k=1}^2 f(x_k^*)\Delta x = f(1)\Delta x + f(2)\Delta x = 1(1) + 4(1) = 5$$

- (c) Approximate the shaded area using a right Riemann sum with $n = 4$ rectangles.

Explanation. $\Delta x = \frac{2}{4} = \frac{1}{2}$

$$x_k^* = x_k = k\Delta x = k\left(\frac{1}{2}\right) = \frac{k}{2}, \text{ for } k = 1, 2, 3, 4$$

$$A \approx \sum_{k=1}^4 f(x_k^*)\Delta x = f\left(\frac{1}{2}\right)\Delta x + f(1)\Delta x + f\left(\frac{3}{2}\right)\Delta x + f(2)\Delta x = \Delta x \left(f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right) + f(2) \right)$$

$$A \approx \frac{1}{2} \left(\frac{1}{4} + 1 + \frac{9}{4} + 4 \right) = \frac{5}{2} + 5 = \frac{15}{2}$$

- (d) Approximate the shaded area using a right Riemann sum with $n = 8$ rectangles.

Explanation. $\Delta x = \frac{2}{8} = \frac{1}{4}$

$$x_k^* = x_k = k\Delta x = k\left(\frac{1}{4}\right) = \frac{k}{4}, \text{ for } k = 1, \dots, 8$$

$$A \approx \sum_{k=1}^8 f(x_k^*)\Delta x = \Delta x \cdot \sum_{k=1}^8 f(x_k^*) = \frac{1}{4} \cdot \sum_{k=1}^8 f\left(\frac{k}{4}\right) = \frac{1}{4} \cdot \sum_{k=1}^8 \left(\frac{k}{4}\right)^2 =$$

$$\frac{1}{4} \cdot \sum_{k=1}^8 \frac{k^2}{16} = \frac{1}{64} \cdot \sum_{k=1}^8 k^2$$

Recall: $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ Therefore

$$A \approx \frac{1}{64} \cdot \sum_{k=1}^8 k^2 = \frac{1}{64} \cdot \frac{8(9)(17)}{6} = \frac{1}{8} \cdot \frac{3(17)}{2} = \frac{51}{16}$$

Problem 4

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Problem 4 A positive continuous function will have area approximated on the interval $[1, 6]$ using n rectangles.

- (a) Find a formula for the grid point, x_k .

Explanation. Here $\Delta x = \frac{b-a}{n} = \frac{5}{n}$.

The basic grid point formula is $x_k = a + k\Delta x = 1 + \frac{5k}{n}$.

- (b) Find a formula for the sample point x_k^* if using a right Riemann sum.

Explanation. For right endpoints, $x_k^* = x_k = 1 + \frac{5k}{n}$.

- (c) Find a formula for the sample point x_k^* if using a left Riemann sum.

Explanation. For left endpoints, $x_k^* = x_{k-1} = 1 + \frac{5(k-1)}{n}$.

- (d) Find a formula for the sample point x_k^* if using a midpoint Riemann sum.

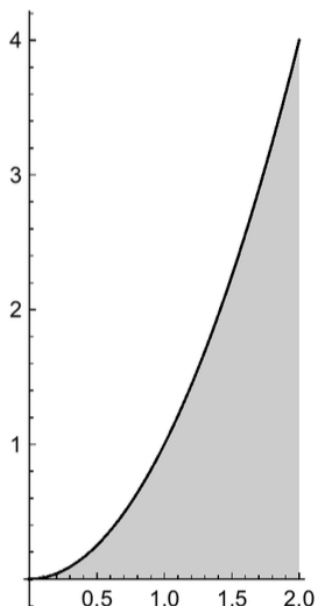
Explanation. For midpoints, $x_k^* = \frac{x_k + x_{k-1}}{2} = 1 + \frac{5(k - \frac{1}{2})}{n}$.

Problem 5

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Problem 5

Problem 5 The graph of the function $f(x) = x^2$ is given in the figure.



Find the exact value of the area of the shaded region. *HINT: Use a right Riemann sum with n rectangles, and then take the limit as $n \rightarrow \infty$.*

Explanation. $\Delta x = \frac{2}{n}$
 $x_k^* = x_k = k\Delta x = k\left(\frac{2}{n}\right) = \frac{2k}{n}$, for $k = 1, \dots, n$

$$A \approx \sum_{k=1}^n f(x_k^*)\Delta x = \Delta x \cdot \sum_{k=1}^n f(x_k^*) = \frac{2}{n} \cdot \sum_{k=1}^n f\left(\frac{2k}{n}\right) = \frac{2}{n} \cdot \sum_{k=1}^n \left(\frac{2k}{n}\right)^2 =$$

$$\frac{2}{n} \cdot \sum_{k=1}^n \frac{4k^2}{n^2} = \frac{8}{n^3} \cdot \sum_{k=1}^n k^2$$

Recall: $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

Therefore

$$A \approx \frac{8}{n^3} \cdot \sum_{k=1}^n k^2 = \frac{8}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{4}{3} \cdot \frac{(n+1)(2n+1)}{n^2} = \frac{4}{3} \cdot \frac{2n^2 + 3n + 1}{n^2}.$$

Let's take the limit of Riemann sums as $n \rightarrow \infty$.

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*)\Delta x = \lim_{n \rightarrow \infty} \frac{4}{3} \cdot \frac{2n^2 + 3n + 1}{n^2} = \lim_{n \rightarrow \infty} \frac{4}{3} \cdot \left(2 + \frac{3}{n} + \frac{1}{n^2}\right) = \frac{8}{3}.$$

Problem 6

February 21, 2026

Problem 6

Problem 6 Consider a Riemann sum with n rectangles for the function f on the interval $[a, b]$.

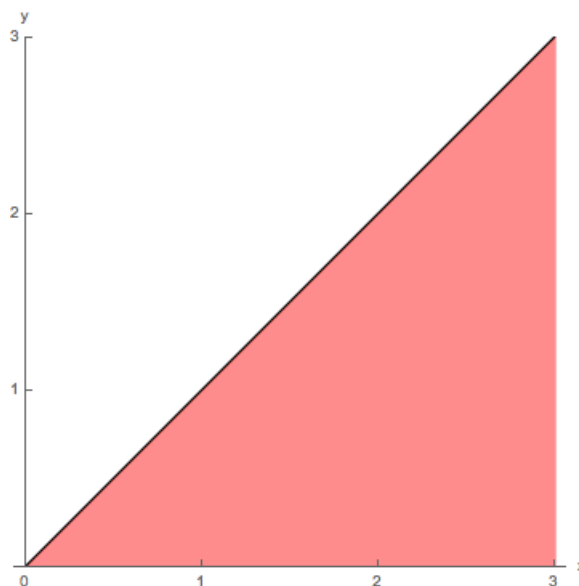
Use geometry to find the limit of Riemann sums as $n \rightarrow \infty$.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x ?$$

(a) $f(x) = x, [0, 3];$

Explanation. $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{k=1}^n x_k^* \Delta x = \frac{9}{2},$

since $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = A$, the area of the shaded region in the figure.

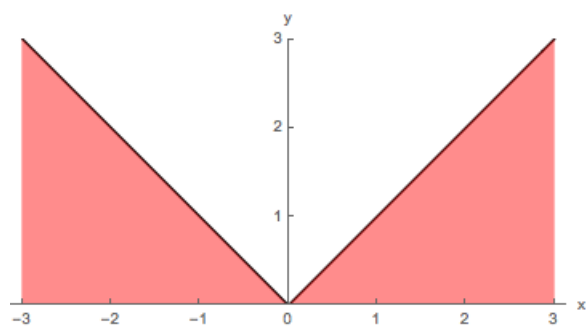


(b) $f(x) = |x|, [-3, 3];$

Explanation. $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{k=1}^n |x_k^*| \Delta x = 9,$

since $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = A$, the area of the shaded region in the figure.

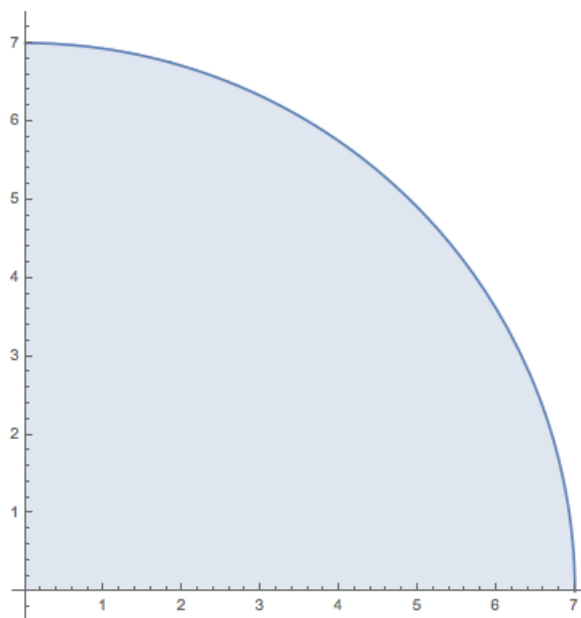
Problem 6



Problem 7

February 21, 2026

Problem 7 A part of a circle is shown in the figure.



- (a) If we express the area of the shaded region in the figure as the limit of Riemann sums

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x,$$

find the function f and the interval $[a, b]$.

Explanation. $f(x) = \sqrt{49 - x^2}$, and the interval $= [0, 7]$.

- (b) Compute the limit.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{49 - (x_k^*)^2} \cdot \frac{7}{n}$$

Explanation. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{49 - (x_k^*)^2} \cdot \frac{7}{n} = \frac{49\pi}{4}$

Problem 8

February 21, 2026

Problem 8 We want to approximate the area under the curve using a right Riemann Sum with the given value of n . Write the sum in summation notation and evaluate it.

(a) $\sin(x)$, $\left[0, \frac{\pi}{2}\right]$, $n = 3$

Explanation. $\Delta x = \frac{b-a}{n} = \frac{\frac{\pi}{2} - 0}{3} = \frac{\pi}{6}$.

$x_k^* = x_k = a + k\Delta x = 0 + \frac{\pi}{6}k = \frac{\pi}{6}k$.

$$\begin{aligned} \sum_{k=1}^3 f(x_k^*)\Delta x &= \sum_{k=1}^3 \left(f\left(\frac{\pi}{6}k\right) \cdot \frac{\pi}{6} \right) \\ &= \frac{\pi}{6} \sum_{k=1}^3 \sin\left(\frac{\pi}{6}k\right) \\ &= \frac{\pi}{6} \left(\sin\left(\frac{\pi}{6}\right) + \sin\left(\frac{2\pi}{6}\right) + \sin\left(\frac{3\pi}{6}\right) \right) \\ &= \frac{\pi}{6} \left(\sin\left(\frac{\pi}{6}\right) + \sin\left(\frac{\pi}{3}\right) + \sin\left(\frac{\pi}{2}\right) \right) \\ &= \frac{\pi}{6} \left(\frac{1}{2} + \frac{\sqrt{3}}{2} + 1 \right) \\ &= \frac{\pi}{6} \left(\frac{3}{2} + \frac{\sqrt{3}}{2} \right) \\ &= \frac{\pi}{12}(3 + \sqrt{3}) \end{aligned}$$

(b) $f(x) = x^2 - 9x + 18$, $[7, 10]$, $n = 6$

Explanation. $\Delta x = \frac{b-a}{n} = \frac{10-7}{6} = \frac{1}{2}$.

$x_k^* = x_k = a + k\Delta x = 7 + \frac{1}{2}k$.

$$\begin{aligned} \sum_{k=1}^6 f(x_k^*)\Delta x &= f\left(7 + \frac{1}{2}\right) \cdot \frac{1}{2} + f(7+1) \cdot \frac{1}{2} + f\left(7 + \frac{3}{2}\right) \cdot \frac{1}{2} + f(7+2) \cdot \frac{1}{2} + f\left(7 + \frac{5}{2}\right) \cdot \frac{1}{2} + f(7+3) \cdot \frac{1}{2} \\ &= f(7.5) \cdot \frac{1}{2} + f(8) \cdot \frac{1}{2} + f(8.5) \cdot \frac{1}{2} + f(9) \cdot \frac{1}{2} + f(9.5) \cdot \frac{1}{2} + f(10) \cdot \frac{1}{2} \\ &= \frac{1}{2} (f(7.5) + f(8) + f(8.5) + f(9) + f(9.5) + f(10)) \\ &= \frac{1}{2} ((7.5^2 - 9(7.5) + 18) + (8^2 - 9(8) + 18) + (8.5^2 - 9(8.5) + 18) + (9^2 - 9(9) + 18) + (9.5^2 - 9(9.5) + 18) + (10^2 - 9(10) + 18)) \\ &= \frac{1}{2} ((7.5^2 + 8^2 + 8.5^2 + 9^2 + 9.5^2 + 10^2) - 9(7.5 + 8 + 8.5 + 9 + 9.5 + 10) + 6(18)) \\ &= \frac{397}{8} \end{aligned}$$

Or in sigma notation:

$$\begin{aligned}
 \sum_{k=1}^6 f(a + k\Delta x)\Delta x &= \sum_{k=1}^6 \left(f\left(7 + \frac{1}{2}k\right) \cdot \frac{1}{2} \right) \\
 &= \frac{1}{2} \sum_{k=1}^6 \left(\left(7 + \frac{1}{2}k\right)^2 - 9\left(7 + \frac{1}{2}k\right) + 18 \right) \\
 &= \frac{1}{2} \sum_{k=1}^6 \left(49 + 7k + \frac{1}{4}k^2 - 63 - \frac{9}{2}k + 18 \right) \\
 &= \frac{1}{2} \sum_{k=1}^6 \left(4 + \frac{5}{2}k + \frac{1}{4}k^2 \right) \\
 &= \frac{1}{2} \left(4 \sum_{k=1}^6 1 + \frac{5}{2} \sum_{k=1}^6 k + \frac{1}{4} \sum_{k=1}^6 k^2 \right) \\
 &= \frac{1}{2} \left(4(6) + \frac{5}{2} \cdot \frac{(6)(7)}{2} + \frac{1}{4} \cdot \frac{(6)(7)(13)}{6} \right) \\
 &= \frac{1}{2} \left(24 + \frac{105}{2} + \frac{91}{4} \right) \\
 &= \frac{1}{2} \cdot \frac{96 + 210 + 91}{4} = \frac{397}{8}
 \end{aligned}$$